

Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments

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Abstract

Forecasting has emerged as a hallmark capability of general-purpose AI, and one where the bar is fundamentally super-human: the accuracy ceiling far exceeds what the best experts achieve. Scientific research itself is a forecasting problem—finite resources mandate exploration/exploitation tradeoff, which in turn relies on prophesying the results of studies before conducting them. We introduce **Foresight-Phys**, a benchmark where AI is asked to predict the results of physical experiments from newest arXiv preprints. In this paper we present the framework along with the results for 135 experiments and 661 typed result fields extracted from 26 physics preprints first released in 2026, after the GPT-5.x training cutoff. The extraction process is automated and is suitable for creating an online benchmark by regularly pulling fresh papers. The strongest model we evaluate, GPT-5.5, attains a mean prediction quality of 87.2% on the benchmark—a soft, per-field score derived from proper scoring rules and designed to reward calibrated forecasts rather than point guesses alone. On numeric quantities it attains a relative Continuous Ranked Probability Score (CRPS normalised by the magnitude of the ground truth for normal targets, and measured in dex for log-normal targets) of 0.41 and places over half (52.4%) of its predictions within one standard deviation of the ground truth. While this indicates nontrivial physical intuition for relevant scales, it also reveals substantial overconfidence compared to an ideal Gaussian observer (68.3%). Code and data are available at <https://anonymous.4open.science/r/Foresight-Phys-7109/>.

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1. Introduction

Forecasting is one of the most difficult problem facing general-purpose AI, demanding the ability to internalize patterns in dynamic environments and reason about consequences in the noisy real world. A large slice of AI benchmarks consist of ever-harder expert-level problem solving (Rein et al., 2023; He et al., 2024; Wang et al., 2026; Zhu et al., 2026). Such benchmarks are gated by what humans can pose and solve, so difficulty saturates as models surpass the graders. Forecasting offers a natural escape: the ground truth is set by reality, not by an examiner. Recent work has emphasized this view for prediction markets and geopolitical events (Halawi et al., 2024; Schoenegger et al., 2024; Karger et al., 2024); we apply it to physical experiments, where the answer is determined by nature and lies at the edge of our collective understanding of the world. The argument is sharpest where intuition pays. A researcher bets a year of grant funding on whether a tweak to a sample, a beamline, or a circuit will move a quantity in the expected direction by the expected order of magnitude. Crucially, he has to balance novelty, which dictates that the result must not come the prior knowledge, with a reasonable chance that the experiment nevertheless yields the desired outcome.

We contribute 1. a paper-derived benchmark of 135 experiments and 661 result fields across five typed answer formats 2. an LLM pipeline for extraction verification, masking, prediction, and scoring—numeric, discrete, and symbolic, the last via an LLM-as-judge 3. results for four frontier OpenAI models evaluated against their respective physical observables

2. Benchmark Design

Paper sourcing. We collected candidate arXiv physics preprints first released between January and May 2026, after the GPT-5.x training cutoff. Domains span condensed-matter ARPES, quantum optics, quantum key distribution, cold atoms, top-quark physics, and topological matter. We use preprints rather than journal versions because the multi-month peer-review lag carries the risk of the results leaking into training corpora. After suitability filtering (below), 26 papers carry usable benchmark questions, yielding 135

experiments and 661 result fields.

Two-pass extraction. Both extraction passes use GPT-5.5 via the OpenAI API with structured output. Pass 1 emits a list of (`experiment_description`, `experiment_results`) tuples per paper. The system prompt requires every sweep limit, apparatus parameter, environmental condition, and unit convention to live in the description, and pushes only *intrinsic*, predictable observables into the targets—explicitly excluding extrinsic artifacts (e.g., the random thickness of a mechanically exfoliated flake) and qualitative claims. Pass 2 re-reads each extracted record and discards experiments unsuitable for blind prediction: descriptions that leak the result, tautological setups, or insufficient context.

Typed answer schema. Each result carries one of five types: `float` (501 fields), `integer` (21), `bool` (73), `categorical` (57, with explicit allowed-value lists), or `formula` (9, symbolic). The typed schema lets us mix numerical proximity, discrete accuracy, and symbolic equivalence in a principled way, and lets one experiment expose multiple independently scorable predictions.

Prediction. For each experiment, every `result` field is recursively replaced with the literal string "TO-PREDICT", while the experiment description, result descriptions, types, units encoded in keys, and allowed categorical values remain visible. The masked JSON is sent with a fixed system prompt and a Pydantic structured-output schema. Numeric fields must return a distribution; boolean fields return `result` and `prob_true`; categorical fields return `result` plus a probability distribution over the allowed values; and formula fields return a symbolic expression plus a confidence.

3. Metrics

The five answer types call for modality-specific proper scoring rules. We give them once, per type, before discussing aggregation.

Numeric (`float`, `integer`). Numeric forecasts are elicited as distributions, not point estimates. For every numeric field the model must output `distribution` $\in \{\text{normal}, \text{log_normal}\}$ and quantiles (p_{10}, p_{50}, p_{90}), with p_{50} used as the point prediction. For `normal` fields, we fit a Gaussian in the target’s original units with $\mu = p_{50}$ and $\sigma = (p_{90} - p_{10}) / (2\Phi^{-1}(0.9)) = (p_{90} - p_{10}) / 2.5631$, then set $z = (y - p_{50}) / \sigma$. For `log_normal` fields, all quantiles and the reference value must be positive; the same fit is performed after applying $\log_{10}$, so σ and the resulting CRPS are measured in dex and $z = (\log_{10} y - \log_{10} p_{50}) / \sigma$.

In both cases the numeric score is the Gaussian CRPS,

$$\text{CRPS} = \sigma \left[z(2\Phi(z) - 1) + 2\phi(z) - \frac{1}{\sqrt{\pi}} \right].$$

Raw CRPS is capped at 30 to keep malformed or catastrophically overconfident forecasts finite. Because raw CRPS carries physical units, the cross-paper metric uses a relative CRPS: $\widetilde{\text{CRPS}} = \text{CRPS}/|y|$ for normal targets with $y \neq 0$, and the dex CRPS directly for `log_normal` targets. Relative CRPS is capped at 3 per field. A normal target with $y = 0$ has no relative CRPS and is omitted from relative-quality averages, although its raw CRPS is still recorded. Calibration is reported through the empirical fractions with $|z| < 1$, $|z| < 2$, and $|z| < 3$.

Boolean. Boolean forecasts return both an `argmax`, `result` $\in \{\text{true}, \text{false}\}$, and a calibrated probability `prob_true`. The exact-match statistic uses the predicted boolean (falling back to `prob_true` ≥ 0.5 if needed). The proper score is the binary Brier score, $\text{Brier} = (p - y)^2$, where $p = \text{Pr}(\text{true})$ and $y \in \{0, 1\}$ is the reference. Boolean quality is $1 - \text{Brier}$; an uninformative $p = 0.5$ forecast has Brier 0.25 and quality 0.75, while the exact-match random-guess baseline is 0.5.

Categorical. Each categorical field carries an explicit allowed-value list of size $k \in \{2, 3, 4, 5\}$ (e.g. *s-p-d-f-wave*; below/at/above the Fermi level). The model returns a best-guess `result` and probabilities over the allowed values. The scorer restricts probabilities to the reference allowed list, clips negative entries to zero, and renormalizes; if the explicit `result` is valid it is used as the `argmax`, otherwise the highest-probability class is used. Exact-match accuracy is recorded. The proper score is the multiclass Brier score $\text{Brier} = \sum_i (p_i - \mathbb{1}_{i=y})^2$, and categorical quality is $\max(0, 1 - \frac{1}{2}\text{Brier})$. Because k varies across questions, the per-question uniform-guess exact accuracy is $1/k_q$; averaging over our k distribution gives a categorical random baseline of 0.293.

Formula. For symbolic fields the prediction is a closed-form expression plus a confidence in $[0, 1]$. Exact equality after whitespace removal is accepted immediately; otherwise the expression is scored by an LLM-as-judge (GPT-5.4-nano) against the reference. The judge receives the experiment description, result description, reference formula, and predicted formula, and is asked to apply a strict equivalence rubric: 1. the prediction must contain every term that appears in the reference; 2. it must contain no spurious extra terms; 3. signs and dimensional prefactors must match; 4. the functional dependence on each input variable must match. Equivalence up to harmless formatting differences, equivalent notation, algebraic rearrangement, and renaming of bound or dummy variables is accepted.

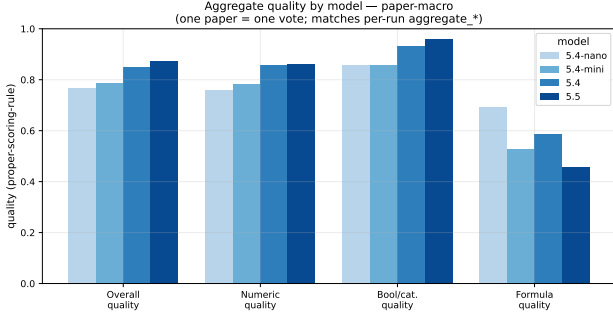


Figure 1. Paper-macro prediction quality by model and modality. GPT-5.5 leads overall at 87.2%, GPT-5.4 follows at 85.0%, and the mini and nano variants lag behind at 78.7% and 76.9%.

Each (experiment, reference, prediction) triple is cached so that re-scoring is deterministic across reruns. Formula accuracy is the fraction judged equivalent. Formula quality uses the same Brier form as booleans, $1 - (c - y)^2$, where c is the model’s confidence and $y = 1$ iff the formula is judged equivalent. The judge is intentionally the weakest model in our lineup, so any judge-induced bias penalises the strongest predictors at least as much as the weakest—a conservative choice for relative comparisons.

Prediction quality. To compare across answer types on a single axis we map every per-field score into a soft *quality* value in $[0, 1]$: $1 - \text{CRPS}/3$ for numeric fields after the relative-CRPS cap, $1 - \text{Brier}$ for boolean and formula fields, and $\max(0, 1 - \frac{1}{2}\text{Brier})$ for categorical fields. This quality axis is what appears in the headline model comparison. Exact-match accuracy is still reported for boolean/categorical argmaxes and for formula equivalence, but quality is the aggregation metric because it rewards calibrated uncertainty: a correct but absurdly overconfident forecast can still be worse than a modest, honest one.

Aggregation. All headline numbers are the paper-macro of the experiment-macro: per-field scores are averaged within each experiment, then within each paper, then across papers. This prevents a single field-rich paper from dominating, and matches how a reader would mentally weight the benchmark—one paper, one vote.

4. Results

Aggregate prediction quality separates the four models (Fig. 1): GPT-5.5 leads at 87.2%, GPT-5.4 follows closely at 85.0%, and the mini and nano variants trail at 78.7% and 76.9% respectively. The overall ranking is consistent with the numeric proper scores and the boolean/categorical quality scores, while the symbolic subset is small enough that we treat it as a diagnostic rather than a stable model-ranking axis.

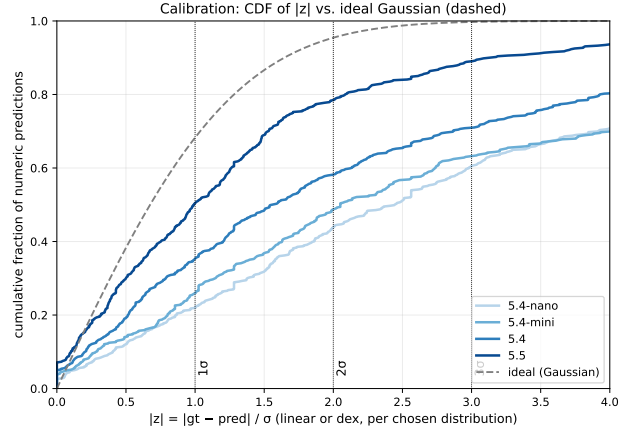


Figure 2. Distribution of prediction errors and coverage on numeric fields, per model. GPT-5.5 captures over half of its numerical predictions within 1σ , though this remains significantly below the 68.3% expected for a calibrated Gaussian distribution.

Numeric. GPT-5.5 places 52.4% of numeric predictions within 1σ , 76.3% within 2σ , and 88.7% within 3σ of ground truth, achieving a mean relative CRPS of 0.413 (raw 3.28 in mixed units) and numeric quality of 86.2%. The strongest two models are close on relative CRPS—GPT-5.5 at 0.413 and GPT-5.4 at 0.423—while the mini and nano variants trail at 0.650 and 0.719. Coverage, however, shows that all models remain overconfident: GPT-5.5’s 1σ coverage is well below the 68.3% expected for a calibrated Gaussian observer, and GPT-5.4, mini, and nano fall further behind at 42.0%, 35.6%, and 29.8% respectively. Thus the models often find the right physical scale, but their uncertainty intervals are still too narrow.

Discrete and symbolic. Boolean and categorical exact accuracy is high across all models: GPT-5.5 reaches 95.8%, GPT-5.4 reaches 92.9%, GPT-5.4-nano reaches 88.6%, and GPT-5.4-mini reaches 83.5%, all comfortably above the 29.3% categorical uniform baseline. Their Brier-derived bool/categorical quality is similarly high, ranging from 85.6% to 95.9%. This indicates that the models often identify the correct qualitative physical regime (topological vs. trivial, gapped vs. gapless, monotonic increase vs. decrease). Formula fields are much noisier: there are only nine symbolic targets, formula exact accuracy ranges from 14.3% to 28.6%, and formula quality ranges from 45.8% to 69.4%. We therefore use the symbolic subset mainly to exercise the extraction and judging machinery, not to draw strong scaling conclusions.

Difficulty profile. Under the analysis definition of “correct” (quality at least 0.5), the current benchmark has a substantial solved mass and a smaller but important tail. Across all 661 fields, 483 (73.1%) are *trivial* (every model

quality-correct), 90 (13.6%) are *easy* (one model wrong), 52 (7.9%) are *discriminative* (split decisions), 18 (2.7%) are *hard* (only one model right), and 18 (2.7%) are *intractable* (no model right). The remaining model-separating tail is dominated by numeric floats: they account for 41 of the 52 discriminative fields and 15 of the 18 intractable fields. This is the benchmark’s main pressure point: qualitative regime recognition is often solved, whereas calibrated continuous physical quantities still expose differences between systems.

5. Related Work

Knowledge-and-recall benchmarks such as ScienceQA (Lu et al., 2022), MMLU (Hendrycks et al., 2021), SciEval (Sun et al., 2024), and C-Eval (Huang et al., 2023) have been complemented by expert-level reasoning benchmarks: GPQA (Rein et al., 2023), OlympiadBench (He et al., 2024), FrontierScience (Wang et al., 2026), and CritPt (Zhu et al., 2026). All are bounded by what human experts can solve and grade; Foresight-Phys instead asks models to forecast outcomes whose ground truth is fixed by physical reality.

The closest prior work is BrainBench (Luo et al., 2025), in which LLMs predict outcomes of neuroscience experiments and exceed domain experts. We adapt the spirit to physics with typed answers, post-cutoff sourcing, and an LLM-driven extraction pipeline. TastyBench (Parv Mahajan, 2025) measures research taste via citation-velocity prediction, a delayed and noisy proxy; Lyu et al. (2025); Park et al. (2026) assess the ability of LLMs for predict the outcome of computational experiments; we target physical observables.

6. Limitations and Future Work

Data leakage. Post-cutoff sourcing mitigates but cannot fully eliminate leakage. It is possible that some works were discussed in public forums before model release, although the nature of fresh preprints makes this less likely than for mature benchmark questions. Embargoed experimental data would be the cleanest solution. A weaker but practical check is post-factum web-search leakage detection over paper titles, arXiv identifiers, and distinctive target phrases.

Paper parsing verification. Parsing experiment definitions and results from papers requires a delicate balance: descriptions must be complete enough to specify the physical system and controls, but must not leak measured values through data-dependent phrasing. For this paper we performed manual spot checks and tracked the fraction of fields all models solve. The current high trivial fraction (483/661) is not by itself evidence of leakage—many boolean and categorical regime questions are genuinely easy—but it is a useful audit signal, and future versions should include inde-

pendent human verification of a random subset of extracted records.

Positivity bias. Published preprints overrepresent successful experiments, so the benchmark inherits a survivorship-biased view of what physics is forecastable. This can be mitigated in three complementary ways: by mining Supporting Materials, Appendices, and PhD theses, where failed experiments are more commonly recorded; by harvesting refutation literature (“Comment on...” and “Reply to...” pieces, failed replications); and by synthetically injecting a predictable flaw into otherwise successful experiments so that the correct prediction is the negative one.

Model Diversity. We restrict our present evaluation to the GPT-5.{4,5} model family. Our primary objective is to introduce and validate the Foresight-Phys benchmarking methodology, rather than to provide an exhaustive survey of all available models. By standardizing on this frontier model family, we demonstrate the benchmark’s ability to differentiate between model tiers and calibrate performance against state-of-the-art reasoning capabilities. The modular architecture of our evaluation harness allows for straightforward expansion to a broader range of open and closed-source models in future work.

Harness. The models in this study rely exclusively on chain-of-thought to make their predictions. In reality, scientists have access to computational tools and external resources. The latter is the harder of the two to grant honestly: unrestricted web access would trivially leak the target. A useful compromise is carefully sandboxed, proxied access to a corpus frozen at the dataset collection start.

7. Conclusion

Forecasting unobserved physical outcomes is a clean, naturally super-human-scaled testbed which pushes foundation models’ forecasting and world-modelling abilities to their limits. Frontier models already achieve meaningful performance: GPT-5.5 reaches 87.2% aggregate quality, while exact boolean/categorical accuracy approaches 96%. Yet the benchmark is not saturated. Numeric forecasts still expose systematic overconfidence—GPT-5.5 reaches only 52.4% 1σ coverage on numeric fields—and the remaining hard tail is dominated by continuous physical quantities. The pipeline, dataset, and per-paper reports are released to grow with each new arXiv month and each new model.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be

specifically highlighted here.

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